

BIOL-1 Comprehensive Final Exam — Answer Key

Part A: Multiple Choice (45 points)

1 point each.

Q	Ans	Brief rationale
1	A	Hierarchy: atom → molecule → cell → tissue → organ (then onward).
2	A	Hypothesis = testable prediction/explanation.
3	C	Homeostasis = stable internal conditions.
4	D	Atomic number = proton count (defines element).
5	B	Covalent = shared electrons.
6	A	Ice lattice via H-bonds → lower density than liquid water.
7	D	Polymer formation releases water: dehydration / condensation synthesis.
8	C	Hereditary information in nucleic acids (DNA/RNA).
9	A	Enzymes (usually proteins) lower activation energy / catalyze specific reactions.
10	D	Cell theory core: life made of cells; cells from cells (modern nuances aside).
11	C	Mitochondria: aerobic respiration / ATP harvest hub in many eukaryotes.
12	A	Prokaryotes lack membrane-bound nucleus (among other differences).
13	D	Fluid mosaic: lipid bilayer + proteins.
14	C	Simple diffusion down concentration gradient without ATP for the crossing itself.
15	B	Osmosis = water across selectively permeable membrane.
16	D	ATP couples energy release to cellular work via phosphate transfer cycles.
17	B	Cellular respiration oxidizes fuel / captures usable energy (often O ₂ -linked overview).

Q	Ans	Brief rationale
18	A	Active site binds substrate; catalyzes reaction.
19	B	Translation = polypeptide synthesis on ribosomes from mRNA.
20	C	DNA polymerase extends new DNA strands during replication.
21	A	RNA: ribose; diverse functional RNAs; typically not archival double helix like nuclear DNA.
22	A	Gametes haploid (n) in humans.
23	D	Crossing over in prophase I of meiosis.
24	C	Mitosis → two genetically identical diploid daughters (barring mutation).
25	C	Alleles = alternative gene versions.
26	B	Complete dominance monohybrid heterozygote cross → 3 : 1 phenotypic ratio (classic).
27	D	Incomplete dominance → intermediate heterozygote; 1:2:1 phenotypes common in textbook cross.
28	C	Epigenetics: gene-use changes without mandatory DNA sequence change.
29	C	Methylation often transcriptionally repressive / “muting.”
30	B	Barr body / X-inactivation = dosage compensation via silencing one X.
31	D	PCR amplifies targeted DNA.
32	A	DNA negative → migrates toward + electrode; smaller fragments often farther.
33	A	CRISPR-Cas9 programmable targeting/cutting at DNA sites (guide RNA).
34	C	Overproduction = many offspring, limited survival/reproduction slots.

Q	Ans	Brief rationale
35	B	Homology = shared ancestry with modified structure/function.
36	A	Fitness = reproductive/genetic contribution to next generations (population genetics sense).
37	B	Gene flow moves alleles between populations via migration/mating.
38	A	Drift strongest when N is small.
39	D	Hardy–Weinberg as null baseline vs observed change.
40	B	Different breeding seasons → prezygotic temporal isolation.
41	D	Sterile hybrid → postzygotic reduced hybrid fertility.
42	D	Geographic split → allopatric speciation pattern.
43	B	Logistic growth slows as N approaches K .
44	C	Disease transmission crowding → density-dependent (typical framing).
45	B	Energy passes through trophic levels with large heat loss; nutrients cycle.

Part B: Fill in the Blank (15 points)

1 point each. Accept close synonyms only where noted.

1. **Homeostasis**
2. **Hydrogen**
3. **Endocytosis** (stem mentions phagocytosis as an example; blank must use the bank term **Endocytosis**)
4. **Lysosome**
5. **Golgi apparatus** (accept Golgi)
6. **Denaturation**

7. **Osmosis**
8. **Mitochondria** (accept singular **Mitochondrion** if student mirrors Part A wording)
9. **Heterozygous**
10. **Codominance**
11. **Epigenetics**
12. **Bottleneck**
13. **Founder**
14. **Hardy–Weinberg** (accept Hardy Weinberg)
15. **Decomposers**

Unused bank distractors: Alleles, Cell wall, Exponential, Translation.

Part C: Integrated Short Answer (25 points)

Student submits any five of seven prompts; 5 points each. Award partial credit.

1. Mitosis vs. meiosis.

- **Mitosis:** one nuclear division (plus cytokinesis) → **2** daughter nuclei/cells; products **diploid** (for typical human somatic starting cell); daughter cells **genetically identical** to each other except mutation; roles include growth/repair/development (asexual reproduction in some organisms).
- **Meiosis:** **two** divisions → **4** haploid products (gametes in animals); **genetically non-identical** owing to crossing over, independent assortment; purpose sexual reproduction / genetic recombination in gametes.

2. Gene → protein + epigenetics.

- **Transcription** (DNA → RNA; nucleus in eukaryotes) → **Translation** (mRNA → polypeptide on ribosomes; cytoplasm/RER) → folding/modifications → functional protein.
- **Epigenetics hook:** promoter methylation dense chromatin or histone modifications can **reduce transcription** without changing DNA letter sequence.

3. Natural selection + analogy vs homology.

- **Selection:** heritable **variation**; **more offspring than environment supports** long-term (overproduction framing); **non-random survival/reproduction** tied to traits → allele-frequency shifts when genetic.
- **Homology:** shared ancestry, structure modified (may differ in function). **Analogy:** similar function/appearance from **convergent** evolution, not necessarily recent common ancestry for that trait.

4. Drift vs. selection + bottleneck + founder.

- **Drift:** random sampling of alleles; magnitude ↑ when **N** small; direction not inherently “for” adaptation.
- **Selection:** systematic differences in survival/reproduction tied to **heritable** traits.
- **Bottleneck:** catastrophe randomly eliminates most individuals; survivors’ allele pool may by chance differ strongly from before.
- **Founder:** small group colonizes new area; allele frequencies reflect **founders’** sampled genes.

5. Exponential vs. logistic + limits.

- **Exponential (J):** per capita growth high early when resources plentiful (idealized unlimited phase).
- **Logistic (S):** growth rate declines as **N** → **K**; **K** = carrying capacity (environment-supported equilibrium-ish ceiling).
- **Density-dependent:** disease spreading with crowding; competition for food when scarce—intensifies with density.
- **Density-independent:** storm/fire/freeze framing that kills similar fraction regardless of density (idealized textbook contrast).

6. Photosynthesis vs. cellular respiration.

- **Photosynthesis:** overview—uses light energy to build sugars from CO₂ and water in producers (chloroplasts in many eukaryotes); outputs O₂ as by-product in oxygenic overview.

- **Cellular respiration:** overview—oxidizes fuel (often glucose) to harvest usable energy with O₂ in many eukaryotes; mitochondria central for aerobic harvest.
- **Relationship:** products of one are substrates/context for the other at ecosystem scale; ATP/NADPH vs ATP from glucose contrasts acceptable at intro level.

7. Species concept + isolation barriers.

- **BSC (intro):** groups interbreeding (or potentially interbreeding) natural populations reproductively isolated from others; fertile/viable offspring framing as taught.
- **Prezygotic example:** temporal / behavioral / habitat / mechanical / gametic isolation blocking mating or fertilization **before** zygote.
- **Postzygotic example:** hybrid inviability, hybrid sterility (e.g., mule), or hybrid breakdown **after** zygote forms.

Part D: Essay (15 points)

One essay chosen. Holistic rubric suggestion: thesis & organization (4), accurate biology across units (7), use of vocabulary (2), conclusion (2). Adjust to course norms.

Option A — Molecules to populations

- DNA stores inherited information in genes; **alleles** are variant forms; **traits** are phenotypic outcomes influenced by genotype + environment + regulation.
- **Mutation** introduces new alleles; **selection** shifts frequencies when traits covary with reproductive success; **drift** randomly shifts frequencies (strong in small populations); **gene flow** homogenizes/diversifies via migration/mating between populations.
- **Populations evolve** = allele/gene-pool frequencies change across generations; individuals carry alleles but evolution is a population-level process over generations.

Option B — Energy, matter, humans

- **Energy** enters as sunlight → producers fix chemical energy; passes **one-way** through consumers with **heat loss** at transfers; not recycled like atoms.

- **Nutrients** (N, P, etc.) move
producer→consumer→**decomposer**→environment→producer cycles.
- **~10% rule** (order-of-magnitude): each step retains only a fraction of production → long chains need **more** primary production per calorie at apex than diets anchored on plants/primary consumers.

Option C — Speciation

- **Biological species concept:** populations interbreeding in nature producing viable fertile offspring within the group; **reproductive isolation** between species-like clusters (limitations acknowledged at AP/intro level).
- **Prezygotic:** habitat/temporal/behavioral/mechanical/gametic examples—block mating/fertilization **before** zygote.
- **Postzygotic:** hybrid inviability / sterility / breakdown—zygote forms but gene flow blocked afterward.
- **Allopatric:** geography splits populations then divergence. **Sympatric:** divergence without geographic isolation—**polyploidy** classic plant example.

End of comprehensive final exam key.